Higher National Diploma in Engineering (Civil)

4th Year, First Semester Examination - 2020

CE 3222 – Foundation Engineering

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 06
Time: 03 hours

Q1).

a) What are the factors to be considered for the selection of subsoil exploration methods? (3marks)

b) What are the information should be included in a subsoil exploration report? (4 marks)

c) State four types of subsoil exploration methods (2 marks)

d) Briefly explain the Standard penetration test procedure? (5 marks)

e) Sub surface of a soil profile revealed that it consists of 6m thick homogenous loose silt sand. The SPT -N value at a depth of 6m from the ground surface (in loose sand layer) was found to be 10. The split spoon sampler was used to collect the soil samples. The length of the drilling rod which is used to connect the split spoon sampler is 10m and no linear effect on the sampler. The diameter of the bore hole was fond to be 120mm. the safety hammer into with energy ratio of 60% used to penetrate the split spoon sampler, estimate the corrected SPT -N value at the depth of 6m below the ground surface. Effective over burden pressure is 205 kPa. Refer Table Q1 for the correction factors. Following equation with usual notations may be useful in the calculations.
(6 marks)

$$C_N = \sqrt{\frac{95.76}{\sigma'_v}}$$

Q2).

a) Using the phase diagram of soil element show that the dry unit weight of the soil sample can be expressed as $\gamma_d = \frac{G_s \gamma_w}{1+wG_s}$

G_s is specific gravity of solid particles, w is the water content and γ_w is unit weight of water. (6 marks)

b) In the natural state, a moist soil has a volume of 0.0093 m^3 and weighs 177.6 N. the oven dry weight of the soil is 153.6 N. If G_s = 2.71. calculate the moisture content (w), moist unit weight (γ), dry unit weight (γ_d), void ratio (e), porosity (n) and degree of Saturation (S). (6 marks)

c) State the objectives of following laboratory tests. (4mrks)

I. Standard proctor test (CBR)

II. Atterberg limit test

III. SPT Test

IV. Sand cone test

d) State four types of pile load testing methods (4mrks)

Q3).

a) Earth retaining structures are categorized as Internally stabilized systems and externally stabilized systems. Explain the functions of these two types of retaining walls by giving two examples for each. (6 marks)

b) A mass concrete retaining wall 6m high. The retained soil material in active state. Draw the lateral earth pressure distribution on the retaining wall and check the stability of the concrete raining wall against overturning. Use the fig Q3, and the details given below. Top layer consists with 3m thick sand layer with a horizontal surface and no surcharge load, soil properties of top layer as C= 0, $\varphi = 35°, \gamma = 16\ kN/m^3$. The bottom layer of sandy clay with 3m thickness C= 15 kN/m^3, $\varphi = 35°, \gamma = 18\ kN/m^3$ and water table is well below the base. assume the density of the concrete as 24 kN/m^3.

(Hint: $\sigma = \gamma z k_a - 2c\sqrt{k_a}$ and , $k_a = tan^2(45 - \frac{\varphi}{2})$) (14 marks)

Q4).

a) Briefly describe the situations in which pile foundations are required. (4marks)

b) Describe what is negative skin friction effect on a pile and explain at which situations this is happening (give three conditions)? (6 marks)

c) A concrete pile is 15m long and 406mm × 406mm in cross section. The pile is fully embedded in sand, for which $\gamma = 17.3$ kN/m^3 and $\varphi = 30°$. following equations are present with the notations with their usual meaning

$Q_p = A_p q' N_q^* \leq A_p q_1$ and $q_1 = 50 N_q^* tan\varphi$

$Q_s = \sum p.\Delta L f$, $f = K.\sigma'_0 tan\delta$ and $L' = 15D$

i. Calculate the ultimate point load using Meyerhof's method (5marks)

ii. Determine the total frictional resistance for K = 1.3 and $\delta = 0.8\varphi$ (5 marks)

Q5).

a) Explain steps of designing of foundation structures (5 marks)

b) State the three failure shear mechanisms of the soil and sketch the Settlement Vs Unit load curves of each of them. (5 marks)

c) Explain the applicability (limitations) of Terzagi's ultimate bearing capacity theory on foundations? (3marks)

d) Determine the Size of square footing to carry net allowable load of 673 KN. The given (factor of safety) FS =3, $D_f = 0.9$m, $\gamma = 18.1$ kN/m^3 , c=0, $\varphi = 40°$ Use Terzagi's equation assuming general shear failure..)Use Table Q5 for the design factors) (7mrks)

Q6).

a) Name and Sketch three types of raft foundations. (4marks)

b) Briefly explain the mechanism and construction sequence of soil nailing technique retaining wall with suitable sketches. (5marks)

c) Explain the meanings of net Ultimate bearing capacity and net allowable bearing capacity of a foundation? (4 marks)

d) A mat foundation measuring 14m× 9m has to be constructed on a saturated clay. For clay, $C_u = 93$ kN/m^3 and $\varphi = 0°$. the depth, D_f for the mat is 2m. Determine the net ultimate bearing capacity. (3marks)

Hint: $q_{(net)} = 5.14 C_u \left(1 + \frac{0.195B}{L}\right)\left(1 + \frac{0.4D_f}{B}\right)$

e) Explain what are compensated foundations? (4 marks)

Table Q1: Correction factors

	Hammer for η_1				**Remarks**
	Donut		Safety		
Country	R-P	Trip	R-P	Trip/Auto	R-P = rope- pulley or cathead
United States					$\eta_1 = {E_r}/{E_{rb}}$
North America	45	-	70 - 80	70 - 100	For U.S trip/auto w/E_r = 80
Japan	67	78	-	-	
United Kingdom	-	-	50	60	$\eta_1 = \frac{80}{70} = 1.14$
China	50	60	-	-	
	Rod Length correction η_2				
Length	>10m		$\eta_2 = 1.00$		N is too high for L < 10m
	6-10m		$\eta_2 = 0.95$		
	4-6m		$\eta_2 = 0.85$		
	0-4m		$\eta_2 = 0.75$		
	Sampler correction η_3				
Without linear		$\eta_3 = 1.00$			Base value N is too high with linear
With linear : dense sand , clay		$\eta_3 = 0.80$			
With linear: loose sand		$\eta_3 = 0.90$			
	Borehole diameter correction η_4				
Hole diameter	60-100mm		$\eta_4 = 1.00$		Base value: N is too small when have an oversize hole
	150mm		$\eta_4 = 1.05$		
	200mm		$\eta_4 = 1.15$		

Table Q5: Bearing capacity factors for shallow foundations

φ (degrees)	N_c	N_q	N_γ
0	5.7	1.0	0.0
5	7.3	1.6	0.14
10	9.6	2.7	1.2
15	12.9	4.4	1.8
20	17.7	7.4	5.0
25	25.1	12.7	9.7
30	37.2	22.5	19.7
35	57.8	41.4	42.4
40	95.7	81.3	100.4
45	172.3	173.3	360.0
50	347.5	415.1	1072.8

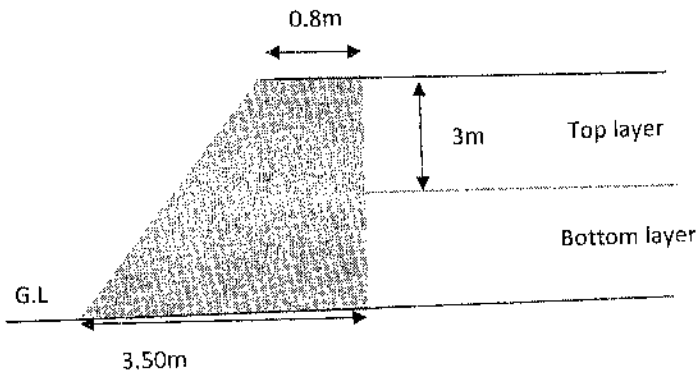

Fig Q3: Mass concrete retaining wall